

7th Grade Unit 3 Part 2 Assessment Guide

General Information	By the end of grade 7, students are expected to perform all four operations with rational numbers with procedural fluency. Units 1 and 2 work together to strengthen a student's understanding of rational numbers. The Unit 3 Part 1 assessment focuses on procedural fluency of operations. The questions on this assessment focus on students applying operations of rational numbers to real-world contexts. The focus moves from the performance of an operation to applying their knowledge of operations. The questions utilize a variety of forms of rational numbers. As 7th grade students often struggle with rational numbers and their equivalent forms, teachers should consider utilizing equivalent forms of rational numbers throughout the course. At the beginning of the assessment is a box with the statements: <i>Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.</i> This suggestion can also help a student self-scaffold a problem. The point distribution is based on these sub-steps. There are students who do not need to show these steps. The point distribution is not meant to punish such students. Students who can think through the problem and find an answer using one or two steps on a calculator such as the Desmos scientific calculator should receive all of the points. It is possible on all of these problems to enter one expression into a calculator such as the Desmos scientific calculator to get the correct answer.
Calculator Information	The questions on this assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 98 points. Teachers can either determine where best to use the extra 2 points or round the total value up to 100 points.

Question	1	Benchmark(s)	MA.7.NSO.2.3								
Deer in Florida Panther National Wildlife Refuge: $16.64 \cdot 41 \frac{1}{3} = 687.78\bar{6}$ Deer in Big Cypress National Preserve: $13.75 \cdot 66 \frac{11}{20} = 915.0625$ Total Deer Population: 687 + 915 or 688 + 915 or 688 + 916 <table><tr><td>Total Deer Population</td><td>1,602, 1,603, or 1,604</td></tr></table>		Total Deer Population	1,602, 1,603, or 1,604	Recommended Scoring (14 Total Points)	Consider distributing the points as: <table><tr><td>Deer in Florida Panther National Wildlife Refuge</td><td>4</td></tr><tr><td>Deer in Big Cypress National Preserve</td><td>4</td></tr><tr><td>Total Deer population</td><td>6</td></tr></table>	Deer in Florida Panther National Wildlife Refuge	4	Deer in Big Cypress National Preserve	4	Total Deer population	6
Total Deer Population	1,602, 1,603, or 1,604										
Deer in Florida Panther National Wildlife Refuge	4										
Deer in Big Cypress National Preserve	4										
Total Deer population	6										
		Additional Information	The student may round in any of the ways shown.								
Question	2	Benchmark(s)	MA.7.NSO.2.3								
$-4.4 - 17\frac{7}{9} \text{ or } -\left(17\frac{7}{9} - (-4.4)\right)$ <table><tr><td>Change in Temperature</td><td>$-\frac{998}{45}$ or $-22\frac{8}{45}^{\circ}\text{C}$</td></tr></table>		Change in Temperature	$-\frac{998}{45}$ or $-22\frac{8}{45}^{\circ}\text{C}$	Recommended Scoring (10 Total Points)	Consider distributing the points as: <table><tr><td>Noting in some form the change in the temperature is the difference between the two locations.</td><td>4</td></tr><tr><td>Change in Temperature</td><td>6</td></tr></table>	Noting in some form the change in the temperature is the difference between the two locations.	4	Change in Temperature	6		
Change in Temperature	$-\frac{998}{45}$ or $-22\frac{8}{45}^{\circ}\text{C}$										
Noting in some form the change in the temperature is the difference between the two locations.	4										
Change in Temperature	6										

Question	3	Benchmark(s)	MA.7.NSO.2.3												
Cabbage: $1\frac{2}{5} \cdot 0.49 = 0.686$ Squash: $2 \cdot 1.29 = 2.58$ Strawberry: $5 \cdot 1.99 = 9.95$ Pepper: $\frac{1}{3} \cdot 0.99 = 0.33$ Total: $0.69 + 2.58 + 9.95 + 0.33$ <table><tr><td>Total Amount</td><td>\$13.55</td></tr></table>		Total Amount	\$13.55	Recommended Scoring (22 Total Points) <table><tr><td>Cabbage</td><td>4</td></tr><tr><td>Squash</td><td>4</td></tr><tr><td>Strawberry</td><td>4</td></tr><tr><td>Pepper</td><td>4</td></tr><tr><td>Total Cost</td><td>6</td></tr></table>	Cabbage	4	Squash	4	Strawberry	4	Pepper	4	Total Cost	6	Consider distributing the points as:
Total Amount	\$13.55														
Cabbage	4														
Squash	4														
Strawberry	4														
Pepper	4														
Total Cost	6														
		Additional Information	The student may round cabbage to 0.69 after the calculation.												
Question	4a	Benchmark(s)	MA.7.NSO.2.3												
April 1: $106.4556 - 28.3231 = 78.1325$ December 1: $106.4556 + 27.7037 = 134.1593$ Difference: $78.1325 - 134.1593$ <table><tr><td>Difference in Stock Price</td><td>−56.0268</td></tr></table>		Difference in Stock Price	−56.0268	Recommended Scoring (14 Total Points) <table><tr><td>April 1</td><td>4</td></tr><tr><td>December</td><td>4</td></tr><tr><td>Difference in Stock Price</td><td>6</td></tr></table>	April 1	4	December	4	Difference in Stock Price	6	Consider distributing the points as:				
Difference in Stock Price	−56.0268														
April 1	4														
December	4														
Difference in Stock Price	6														
		Additional Information	The student should not round based on the information given, but if they consider price as having 2 decimal places then consider taking off 1 point. 56.0268 should also be accepted as a correct answer since order wasn't specified.												

Question	4b	Benchmark(s)	MA.7.NSO.2.3						
January 2: $106.4556 - 5.9759 = 100.4797$ Trevor's stock: $1000 \cdot 100.4797$ Cost of 1,000 shares on January 2, 2020 100,479.7		Recommended Scoring (10 Total Points) Additional Information	Consider distributing the points as:<table><tr><td>January 2</td><td>4</td></tr><tr><td>Cost</td><td>6</td></tr></table> The student may round in any of the ways shown. Students may show all work in one expression.	January 2	4	Cost	6		
January 2	4								
Cost	6								
Question	5	Benchmark(s)	MA.7.NSO.2.3						
$872\frac{5}{12} - 650 = 222.41\bar{6}$ $222.41\bar{6}$ $50\frac{3}{5}$ Time 4.4 minutes		Recommended Scoring (14 Total Points) Additional Information	Consider distributing the points as:<table><tr><td>Difference in current position and safe depth</td><td>4</td></tr><tr><td>Difference divided by rate</td><td>4</td></tr><tr><td>Time</td><td>6</td></tr></table> The student may show the difference between the two depths in a variety of ways. Students should not lose points if the difference is written as a negative in their work. Consider deducting 2 points if the student gives time as a negative value.	Difference in current position and safe depth	4	Difference divided by rate	4	Time	6
Difference in current position and safe depth	4								
Difference divided by rate	4								
Time	6								

Question	6	Benchmark(s)	MA.7.NSO.2.3						
2001: $1167.6 \cdot 174\frac{5}{12} = 203648.9$ 2019: $1167.6 \cdot 107\frac{3}{32} = 125042.6625$ Difference: $203648.9 - 125042.6625$		Recommended Scoring (14 Total Points)	Consider distributing the points as: <table><tr><td>2001</td><td>4</td></tr><tr><td>2019</td><td>4</td></tr><tr><td>Difference</td><td>6</td></tr></table>	2001	4	2019	4	Difference	6
2001	4								
2019	4								
Difference	6								
Difference in the Average Emissions of CO₂	78,606.2375 grams	Additional Information	Students may first find the difference in the monthly average and then multiply by the average distance traveled by month.						